Feature Importance for Applied Research

From model inspection to statistical inference

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Why feature importance?

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> **!** “Importance” is not a single quantity → *method choice* defines the answer

Which features matter?

Example: Predicting bike rentals

Term	Estimate	SE	p
`hour`	7.8	0.4	< 0.001
`humidity`	−197.6	16.4	< 0.001
`temperature`	350.8	22.5	< 0.001
`windspeed`	18.9	22.6	0.403

`lm(count ~ .)` on bike sharing data (more later)

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📝 **But**: Important *for what*, and *how* is it measured?

Importance ≠ effect

Effect

- "What happens to $\hat{y}$ *when* X_j *changes*?"
- Linear model: β_j
- *Signed*, in units of X_j

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- Scale depends on importance method and loss

  → Usually read as a *ranking*

Feature Importance Estimands

A rough taxonomy of interpretibility methods

Today: model-agnostic, loss-based, global

Model-specific

(G)LM coefficients, Gini impurity, gradient-based

|

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Local
One observation's prediction (e.g., SHAP)

vs.

Global
the model as a whole, feature-specific

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Which question?

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  Is X_j related to Y at all?

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⚠ Popular methods estimate *model* importance first, research questions often target *data*

Feature Importance Methods

Why not just refit?

Leave-one-covariate-out (LOCO) (Lei et al. 2018)

Calculation

1. Fit model $\hat{f}$ with all features
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📝 **Test loss, e.g. mean squared error**

$L\left(\hat{f}\right) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$ → importance is on the scale of the *loss*

Don’t refit, perturb

Permutation feature importance (PFI) (Breiman 2001)

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1. Fit model $\hat{f}$ once
2. *Shuffle* X_j in the test data: $\tilde{x}_j$
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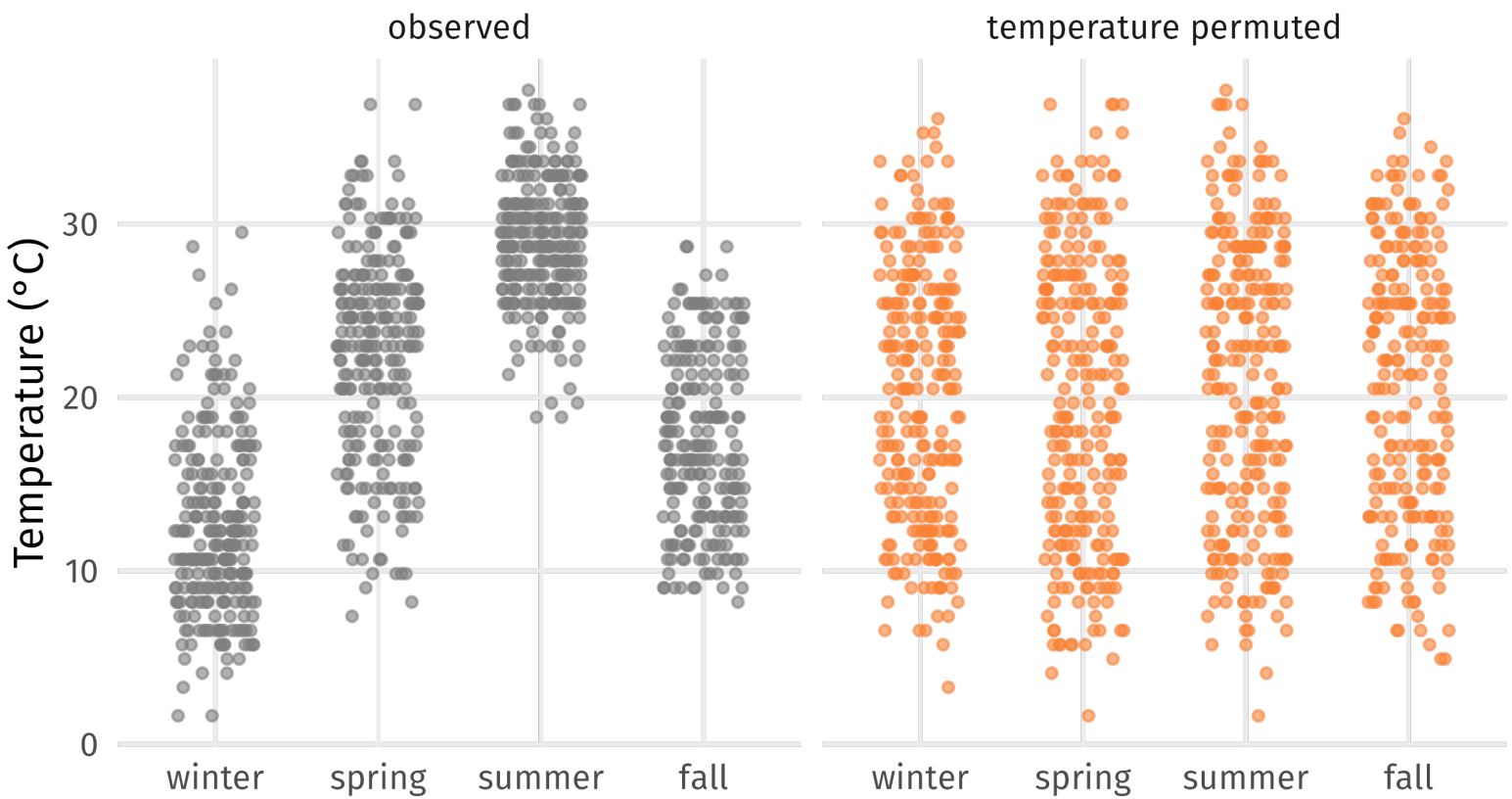
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→ A 20-year-old patient with 30 years of smoking

Shuffling leaves the data

Model is evaluated where it has never seen data (Hooker et al. 2021)



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! Needs samples $\tilde{x}_j$ from $X_j \mid X_{-j}$, which are not easy to get

Where do conditional samples come from?

- *Gaussian*: fast, continuous features only
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📝 **Adversarial random forests (ARF)**

- Generative model for tabular data, estimates conditional density (Watson et al. 2023)
- Categorical features, missing values
- Default sampler for CFI in `{xplainfi}`

Between PFI and CFI

Relative feature importance (RFI) (König et al. 2021)

Sample X_j from $X_j \mid X_G$ for a chosen *conditioning set* G

$G = \varnothing$

- PFI
- "Does the model use X_j?"

$G = \{\text{age}, \text{sex}\}$

- RFI
- "Beyond age and sex?"

$G = -j$

- CFI
- "Beyond all other features?"

Is it more than noise?

Inference for CFI

- CFI is a mean of per-observation loss differences, $\tilde{x}_i$: x_i with x_{ij} perturbed

$$\Delta_i = \left(y_i - \hat{f}(\tilde{x}_i)\right)^2 - \left(y_i - \hat{f}(x_i)\right)^2, \quad \text{CFI}_j = \overline{\Delta} = \frac{1}{n}\sum_{i=1}^{n} \Delta_i$$

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- One-sided paired test on $\overline{\Delta}$ (e.g., t-test, Wilcoxon)
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💡 Holdout set, model fixed → *model* importance

And now for some game theory

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- **Shapley value**: average gain from adding j (Shapley 1953)

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📝 **SAGE vs. SHAP**

SAGE: *global*, explains the *loss* for features (Covert et al. 2020)

SHAP: *local*, explains *prediction* for observations (Lundberg and Lee 2017)

Application Example

The bike sharing dataset

Capital Bikeshare, Washington, D.C., 2011–2012

- *Outcome*: bike rentals per hour
- 3000 samples from ~17k
- Gradient boosting (`xgboost`, pre-tuned) and `lm`
- Holdout: 80% train, 20% test

9 features

- **Calendar**: hour, weekday, working day, season, year
- **Weather**: temperature, humidity, wind speed, weather type

Setup for code example

{mlr3} objects, defined once


```
library(xplainfi)
library(mlr3learners)
library(mlr3pipelines)

task <- tsk("bike_sharing")
learner <- as_learner(
  po("encode") %>>%
    lrn("regr.xgboost", nrounds = 4932, eta = 0.0023, max_depth = 8)
)
measure <- msr("regr.mse")
resampling <- rsmp("holdout", ratio = 0.8)
```

PFI


```
pfi <- PFI$new(
  task, learner, measure,
  resampling = resampling,
  n_repeats = 10
)
pfi$compute()
pfi$importance()
```

Feature	PFI
hour	43478
temperature	8526
working_day	6532
year	6120
humidity	1983
season	1365
weekday	912
weather	673
windspeed	98

CFI


```
cfi <- CFI$new(
  task, learner, measure,
  resampling = resampling,
  n_repeats = 10,
  sampler = ConditionalARFSampler$new(task)
)
cfi$compute()
cfi$importance()
```

Feature	CFI
hour	36061
year	5537
temperature	4817
working_day	3268
humidity	1180
season	1020
weekday	735
weather	567
windspeed	89

LOCO


```
loco <- LOCO$new(
  task, learner, measure,
  resampling = resampling
)
loco$compute()
loco$importance()
```

Feature	LOCO
hour	20460
year	3285
temperature	1253
working_day	646
season	542
weather	331
weekday	239
humidity	150
windspeed	-32

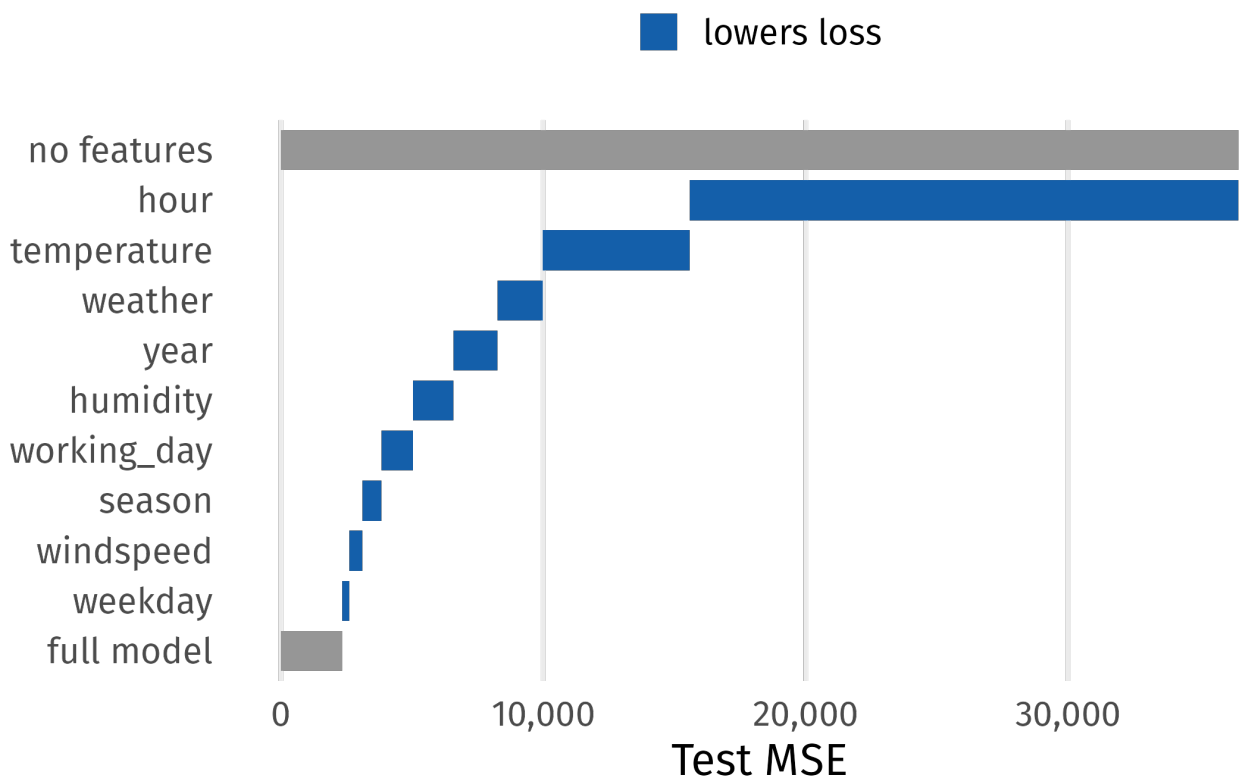
SAGE


```
sage <- MarginalSAGE$new(
  task, learner, measure,
  resampling = resampling,
  estimator = "kernel",
  n_coalitions = 16384
)
sage$compute()
sage$importance()
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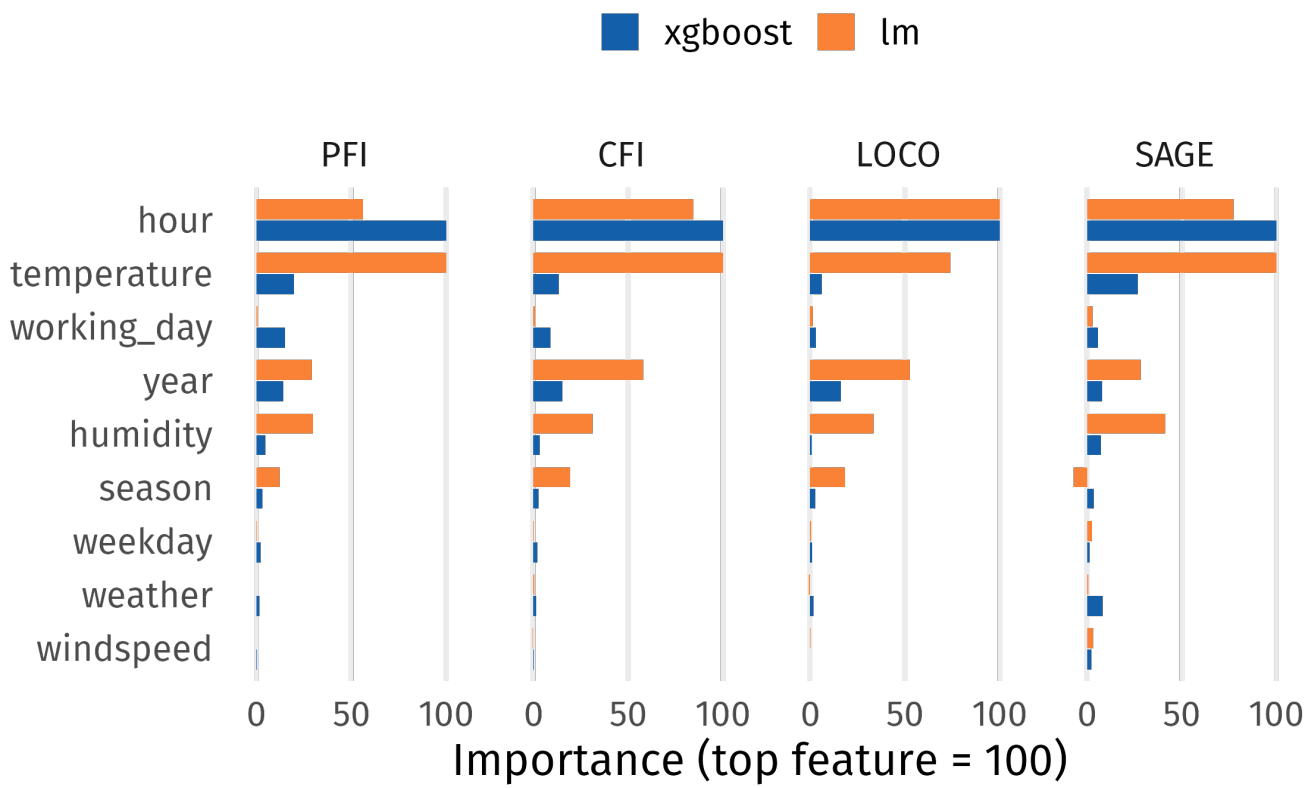
Feature	SAGE
hour	20923
temperature	5593
weather	1731
year	1687
humidity	1533
working_day	1197
season	725
windspeed	499
weekday	274

SAGE adds up

Loss without features → loss of the model, one step per feature



Comparing methods and learners



Inference

CFI with CPI, 600 held-out hours


```
cfi$importance(ci_method = "cpi", p_adjust = "BH")
```

Feature	CFI	SE	Lower 95%	p (BH)
hour	36061	2035	32064	< 0.001
year	5537	491	4572	< 0.001
temperature	4817	541	3754	< 0.001
working_day	3268	349	2582	< 0.001
humidity	1180	239	710	< 0.001
season	1020	146	733	< 0.001
weekday	735	99	540	< 0.001
weather	567	122	328	< 0.001
windspeed	89	66	-40	0.175

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working_day	3268	349	2582	< 0.001
humidity	1180	239	710	< 0.001
season	1020	146	733	< 0.001
weekday	735	99	540	< 0.001
weather	567	122	328	< 0.001
windspeed	89	66	-40	0.175

- `hour`, `year`, `temperature` clearly important
- `windspeed`: no evidence after BH adjustment

Summary

Question	Method	Watch out for
Does the model use X_j?	PFI	extrapolation
Information beyond the others?	CFI: perturb	conditional sampler
	LOCO: refit	refits, learner tuning
Share of the explained loss?	SAGE	cost, estimator noise

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- CFI and LOCO: near-identical estimands, different estimation (Ewald et al. 2024)
- Generalize to learner/DGP importance: Combine with resampling procedure
  → inference becomes more complicated (e.g., Nadeau-Bengio correction)

Where to go next

1. `install.packages("xplainfi")`
2. Documentation: https://mlr-org.github.io/xplainfi
3. `{xplainfi}` and practical guidance:

   xplainfi: Feature Importance and Statistical Inference for Machine Learning in R

   Burk et al. (2026), The R Journal
4. Methodological background:

   A Guide to Feature Importance Methods for Scientific Inference

   Ewald et al. (2024), Explainable Artificial Intelligence (xAI 2024)

Thank you for your attention!

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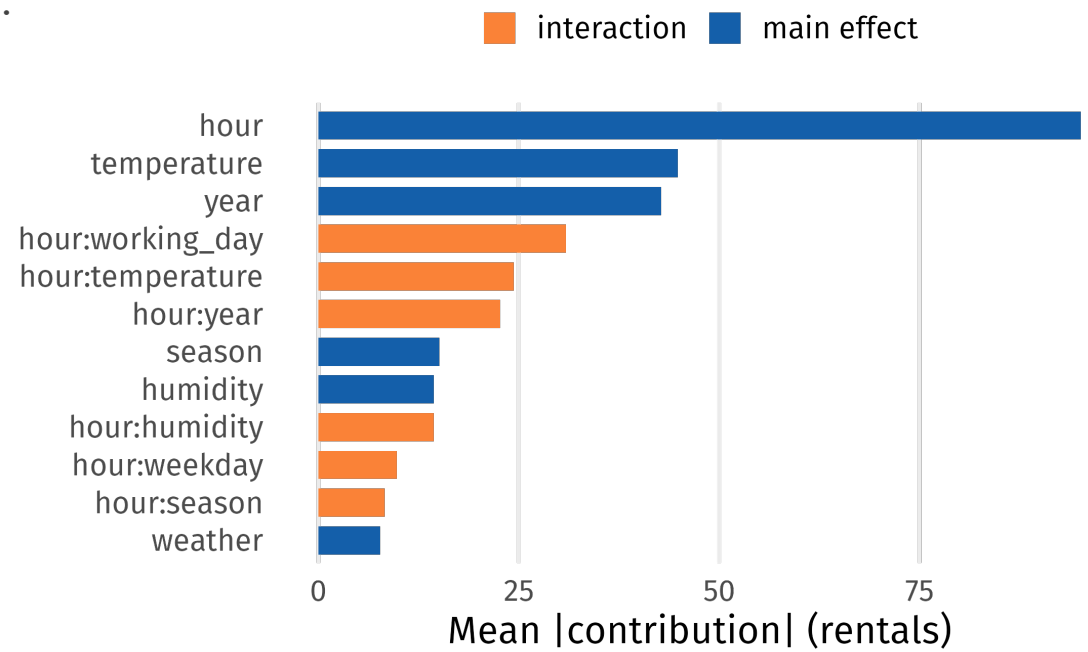
Backup

Prediction-based: decomposing $\hat{f}$

Functional decomposition of the `xgboost` model with `glex` (Hiabu et al. 2023)

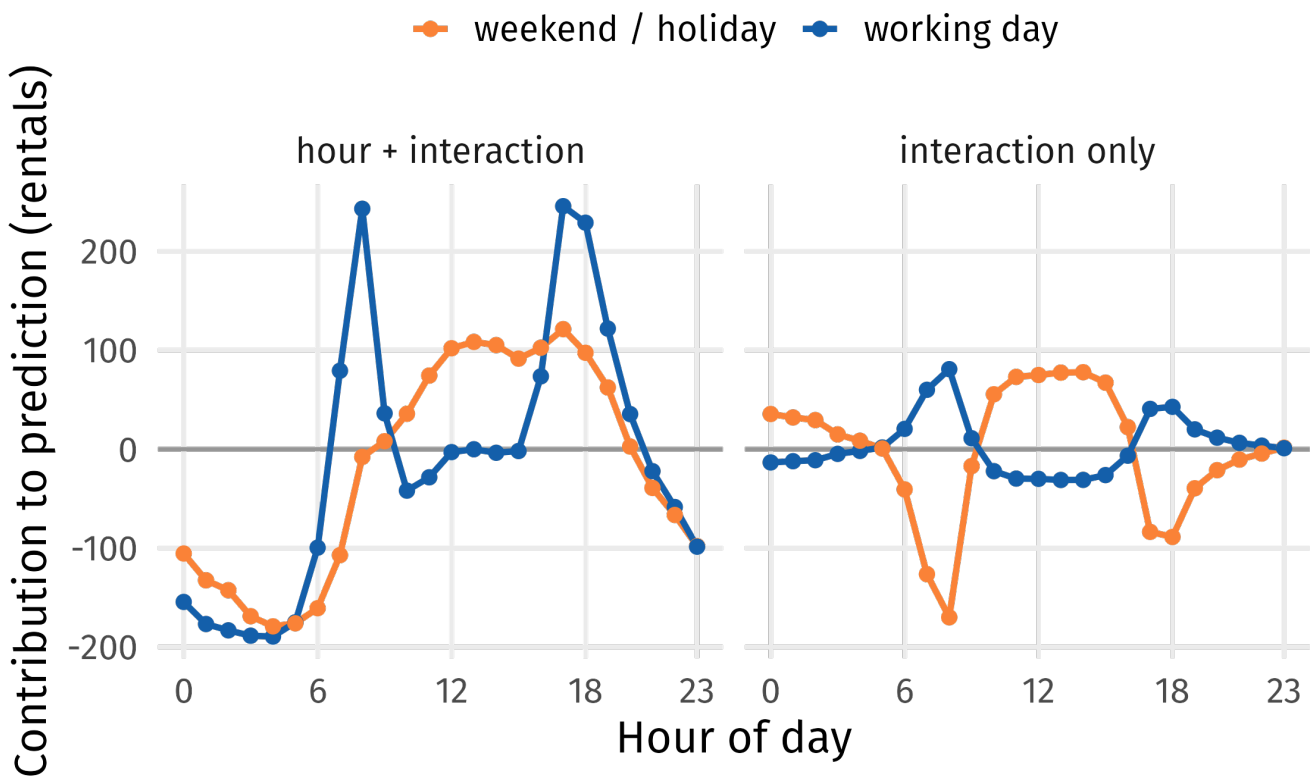
$$\hat{f}(x) = m_0 + \sum_j m_j(x_j) + \sum_{j<k} m_{jk}(x_j, x_k) + \dots$$

- Components on the scale of the *prediction* (rentals), not the loss
- Importance of a component: mean $|m_S|$
- No refits, no sampling: read off the trees



Low importance ≠ no effect

`working_day` matters through its interaction with `hour`



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